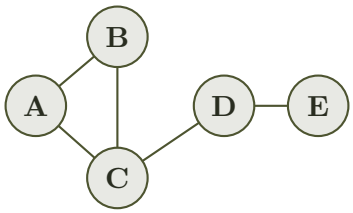
Module 8: Graph Theory and Networks — Summary

Central idea: The arrangement of connections determines who can reach whom and whether a system can continue working after a failure.



A **node** (vertex) is a place or person. An **edge** is a direct connection. The graph above has nodes A–E and edges AB, AC, BC, CD, DE. A line that passes near a node without meeting it does not create a connection. The drawing is not a map to scale.

Count direct neighbors

The **degree** of C is 3, because it touches A, B, and D. Degrees A–E are 2, 2, 3, 2, 1; their sum is 10. Each of the five edges gets counted at two ends: sum of degrees $= 2 \times$ number of edges.

A **path** travels along existing edges. A–C–D–E uses **three edges**, so its length is 3. A **shortest path** uses the fewest edges when links are unweighted. A and E are connected through this path even though they are not direct neighbors. If edges have travel times, add those times instead of counting links.

A **connected component** is a group in which any node can reach any other. The original graph has one component. If C–D fails, the components are $\{A, B, C\}$ and $\{D, E\}$. The edge C–D is a **bridge**, because removing it separates the graph. By contrast A–B is not a bridge: A and B can still reach each other through C.

Module 8: Graph Theory and Networks — Summary (continued)

Use the mathematics to compare choices.

One new link, different outcomes

Suppose C–D might fail. Adding A–C would not help: that link already exists. Adding B–E creates a new route B–E between the two sides. After C–D fails, A–B–E–D still works. The **location** of the new edge matters more than the count alone.

Begin with A informed in round 0. In each round, informed nodes send to their direct neighbors. Recipients can relay in the *next* round; all direct links work.

Round	Newly informed	Total informed
0	A	A
1	B, C	A, B, C
2	D	A, B, C, D
3	E	A, B, C, D, E

With this rule, the first round a node receives an alert equals the shortest-path distance from A. A failed bridge can prevent delivery entirely.

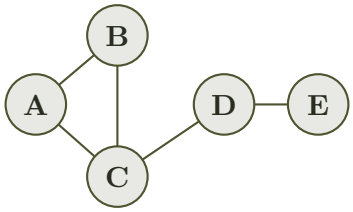
1. Say what nodes and edges mean in the situation.
2. Decide whether the question concerns direct contacts, routes, failure, or spreading.
3. Count neighbors or trace paths; verify every link exists.
4. Compare the original network with one specific change.
5. Explain a limitation (such as travel time, unreliable contacts, or accessibility).

Remember: Two networks can have the same number of edges and very different routes and vulnerabilities. The arrangement matters.

Module 8: Graph Theory and Networks — Prequiz

Open-note practice. Show your counts and routes. Brief explanations are enough.

Name: _____ Date: _____



Use the graph above for all five questions. Edges: AB, AC, BC, CD, DE.

1. What are C's direct neighbors? Find the degree of C. Check the handshake rule by adding all five degrees.

2. Find one shortest path from A to E. Write its nodes in order and give its length in edges. Explain why A and E are connected even though no direct edge joins them.

3. Suppose C–D fails. List the connected components. Is E reachable from A? Explain briefly.

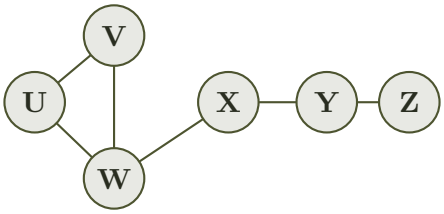
4. The town can add just one connection: A–E or B–D. Both choices preserve a connection between A and E after C–D fails. Trace a surviving path for each choice. Which one gives the shorter A-to-E path?

5. Begin with only A informed in round 0. Every informed node alerts its direct neighbors in the next round. Newly informed nodes relay one round later. List the **newly informed nodes** in rounds 1, 2, and 3 and the **total number informed** after round 2.

Module 8: Graph Theory and Networks — Check-In

Same core skills as the prequiz, with a different community network. Show enough work to make your thinking visible.

Name: _____ Date: _____



The nodes U, V, W, X, Y, Z represent community centers and the edges are UV, UW, VW, WX, XY, YZ.

1. List W's direct neighbors and degree. Find the sum of all six node degrees, and check it against the number of edges.

2. Give one shortest path from U to Z and its length in edges. Explain why U can reach Z without a direct U–Z connection.

3. Remove W–X. List the resulting connected components. Can U still reach Z? Give a reason.

4. A grant allows one additional link: U–X or X–Z. Which proposal keeps U connected to Z if W–X later fails? Give a surviving path and explain why the other proposal does not work.

5. Start with only X informed in round 0. Informed locations send to neighbors each round; new recipients relay the following round. List the **newly informed nodes** in rounds 1 and 2. How many locations are informed in total after round 2?